

BRIGHTLEARN DATA ANALYTICS

SQL FUNDAMENTALS EXERCISE 2

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Course: Brightlearn Basic SQL

SQL Aggregate Functions and Operators.

SECTION 1

Q1. Students

student_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

Q1. All distinct departments in the students table

```
SELECT DISTINCT department  
FROM students;
```

department
IT
HR
Finance

Q2. The average age of students per department.

```
SELECT department, AVG(age) AS avg-age  
FROM students
```

GROUP BY department;



IT: $(20 + 21) \div 2 = 20.5$

HR: $(22 + 22) \div 2 = 22$

Finance: $23 \div 1 = 23$

department	avg-age
IT	20.5
HR	22
Finance	23

Q3. Departments with more than 1 student.

SELECT department, COUNT(student_id) AS student_count
FROM students

GROUP BY department
HAVING COUNT(student_id) > 1;

department	student-count
IT	2
HR	2

Q4. All students whose age is between 21 and 23

SELECT student_id, name, age, department
FROM student

WHERE age BETWEEN 21 AND 23;

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

Q5. All students in the IT or HR department who are older than 21

SELECT student_id, name, age, department
FROM students

WHERE department IN ('IT', 'HR')
AND age > 21;

student_id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

SECTION 2

Courses

course-id	Course-name	department	Credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

Q6. Total Credits per department, Only for departments with more than 5 total credits.

```
SELECT department, SUM(Credits) AS total_credits
FROM courses
GROUP BY department
HAVING SUM(Credits) > 5;
```

department	total_credits
IT	11

Q7. All courses that do not have 4 credits

```
SELECT course_id, course_name, department, credits
FROM courses
WHERE credits != 4;
```

course-id	Course-name	department	Credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

Q8. The top 3 courses by credits in descending order.

```
SELECT course_id, course_name, Credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

SECTION 3

Q9. enrollments

enrollment_id	student_id	Course_id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

Q9. Get the Maximum, minimum and average grade across all enrollments.

```
SELECT MAX (grade) AS max-grade,
       MIN (grade) AS min-grade,
       AVG (grade) AS avg-grade
FROM enrollments;
```

max-grade	min-grade	avg-grade
90	78	84.6

Q10. Count how many enrollments exist per course.

SELECT course_id, COUNT(enrollment_id) AS enrollment_count
FROM enrollments
GROUP BY course_id;

Course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

Q11. SECTION 04

Salaries

employee_id	name	department	Salary	bonus
1	Tom	IT	60000	5000
2	Jerry	HR	55000	4000
3	Spike	Finance	70000	6000
4	Tyke	IT	62000	5500
5	Butch	HR	54000	3500

Q11 Total salary and total bonus per department

SELECT department,
SUM(salary) AS total_salary,
SUM(bonus) AS total_bonus

FROM salaries

GROUP BY department;

IT: Salary: $60000 + 62000 = 122000$
 Bonus: $5000 + 5500 = 10500$
 HR: Salary: $55000 + 54000 = 109000$
 Bonus: $4000 + 3500 = 7500$
 Finance: Salary: 70000
 Bonus: 6000

department	total_salary	total_bonus
IT	122000	10500
HR	109000	7500
Finance	70000	6000

Q12 SHOW departments where average salary is above 55000

```
SELECT department, AVG(salary) AS avg_salary  
FROM salaries  
GROUP BY department  
HAVING AVG(salary) > 55000
```

department	avg_salary
IT	61000
Finance	70000

Q13 Employees whose salary plus bonus is greater than 60000

```
SELECT employee_id, name, salary
```

```
SELECT employee_id,  
name,  
salary,  
bonus,
```

```
salary + bonus AS total_compensation
```

```
FROM salaries
```

```
WHERE salary + bonus > 60000
```

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

SECTION 05

Projects

Project-id	Project-name	department	budget
1	AI App	IT	120000
2	Payroll system	Finance	80000
3	Dashboard	IT	150000
4	Website	Marketing	60000
5	HR Portal	HR	50000

Q 14. Total and average budget per department. Only include departments with average budget above 70000

```
SELECT department,  
       SUM(budget) AS total-budget,  
       AVG(budget) AS totalavg-budget  
FROM projects  
GROUP BY department  
HAVING AVG(budget) > 70000;
```

department	total-budget	avg-budget
IT	270000	135000
Finance	80000	80000

Q 15. All projects with budgets between 50000 and 120000, excluding the Market department

```
SELECT project-id, project-name, department, budget  
FROM projects  
WHERE budget BETWEEN 50000 AND 120000  
AND department != 'Marketing';
```

Project-id	Project-name	department	budget
1	AI App	IT	120000
2	Payroll System	Finance	80000
5	HR Portal	HR	50000